

# MESHARY ASDALA AQUINO

Full-Stack Software Engineer | React, Next.js, Python, PostgreSQL  
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## SUMMARY

Full-stack engineer with 2 years building web applications end to end: React and Next.js on the front, FastAPI and PostgreSQL behind it, most of it as the sole engineer on the system. Ships the whole path from relational schema through REST API, authentication and role-based access, responsive interface, and deployment. Also builds LLM-backed features under production constraints, including schema-constrained structured output and confidence-gated human review. Open to fully remote roles with overlap into US and European hours.

## PROFESSIONAL EXPERIENCE

**Software Engineer, Independent Contract** | Remote | September 2024 - Present (part-time, concurrent)

- Built real-time video and messaging for LearnMate, a cross-platform tutoring app used by WMSU undergraduate researchers, designing the Firebase data model, role-based auth, and the WebRTC signaling layer. (Dart, Flutter, Firebase)
- Delivered on-device wood knot detection as sole engineer for AdZU researchers by training a YOLOv8 detector on a 12,000-image dataset and shipping the ONNX-to-TFLite quantization and inference pipeline for real-time mobile use. (Python, ONNX, TensorFlow Lite)

**IT Specialist** | M. Abubakar Construction and Engineering | Zamboanga City, Philippines | June 2025 - June 2026

- Cut document and certificate reconciliation from three days to under four hours by replacing a shared-spreadsheet process with a normalized PostgreSQL tracking system, authenticated across three user tiers with role-based access control.
- Eliminated 15+ hours of weekly manual data entry across a 50-employee payroll cycle by building a Python desktop application (PyQt6) that automated statutory deductions, contribution computations, and payslip generation end to end.

**Software Developer Intern** | Regional Trial Court, Region 9 | Zamboanga City, Philippines | August 2024 - April 2025

- Kept court operations running for 10+ daily staff through repeated network outages by building an offline-first desktop client with automated cloud synchronization.
- Reconciled 1,000+ legal records after outages with zero data loss by engineering bi-directional sync with conflict resolution between offline clients and the live database.

## PROJECTS

**Enterprise Payroll Web Application** | Next.js 14, React 18, TypeScript, FastAPI, PostgreSQL, TailwindCSS | [github.com/Meshy21/payroll-online-web](https://github.com/Meshy21/payroll-online-web)

- Delivered BIR TRAIN Law-compliant payroll across semi-monthly, weekly, and daily frequencies by building a tax engine implementing the 2023 revised schedules with 2025 SSS, PhilHealth, and Pag-IBIG contribution tables.
- Secured employee and bank data with AES-256 field encryption, JWT auth with bcrypt hashing, RBAC on every protected REST endpoint, and an immutable audit trail.

**AI Enquiry Triage and Routing Service** | Python, Google Gemini API, Pydantic, Streamlit | [github.com/Meshy21/strata-ai-router](https://github.com/Meshy21/strata-ai-router)

- Eliminated response-parsing failures in an LLM classification pipeline by constraining model output to a Pydantic schema (category, confidence, action, draft reply) instead of parsing free text, producing JSON consumable directly by downstream CRM and webhook integrations.
- Designed for human-in-the-loop safety: classifications below a confidence threshold route to manual review instead of auto-replying, and vague or adversarial input is classified into an explicit spam class rather than answered.

**LLM Email and Document Automation Pipelines** | n8n, OpenAI API, Google Gemini multimodal, Google Workspace APIs

- Automated inbound Gmail triage end to end: LLM intent classification drives label routing, sender logging to Sheets, and parallel attachment archival to Drive.
- Built a receipt pipeline intercepting photos via Telegram webhook, using multimodal vision at temperature 0 to extract merchant, amount, and date into structured expense rows with timestamped audit archival.

**SyncSolve API, stateless conflict-resolution engine** | Python, FastAPI, Docker, Kubernetes, Prometheus | [github.com/Meshy21/SyncSolve-API](https://github.com/Meshy21/SyncSolve-API)

- Resolves offline-first sync conflicts at field and document level using vector-clock causality, returning HTTP 409 with RFC 6902 JSON Patch deltas and the exact conflicting dot-paths so clients merge without data loss.
- Instrumented for production: pytest suite, OpenAPI docs, structured JSON logging with request correlation IDs, Prometheus request and conflict metrics, Dockerfile and Kubernetes manifests.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Dart, SQL, PHP, HTML/CSS

**Frontend:** React 18, Next.js 14, TailwindCSS, Vite, responsive UI, Flutter, PyQt6, WebRTC

**Backend and Data:** FastAPI, PostgreSQL, MySQL, SQL Server, Firebase, REST API design, relational schema design, JWT, RBAC, AES-256 field encryption

**Infrastructure:** Docker, Kubernetes, Git/GitHub, pytest, OpenAPI/Swagger, Prometheus, structured logging, Linux, Agile/Scrum

**AI and LLM:** OpenAI API, Google Gemini API, schema-constrained structured output (Pydantic), prompt design, multimodal extraction, n8n workflow automation

**ML and Vision:** YOLOv8, ONNX, TensorFlow Lite, OpenCV, dataset training, quantization, on-device inference

## EDUCATION & CERTIFICATIONS

Mapua University, MEng Computer Engineering (in progress) | 2026 - Present

Western Mindanao State University, BS Computer Engineering | 2021 - May 2025 | GPA 1.71 (1.00 highest, first-class range)

Data Associate Pathway (DAP), 2025 | Cisco Networking Basics and Python Essentials 1, 2025 | Civil Service Professional Eligibility (CSC)